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Beyond Language Priors: Diagnosing and Fixing Visual-Origin Hallucinations in Multimodal LLMs

Cosine Similarity and Grad-CAM Entropy Reveal That Half of Object Hallucinations Originate in the Visual Encoder, Not the Language Model

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Object hallucination in multimodal large language models (MLLMs) — the generation of text describing objects not present in the image — is a persistent failure mode that undermines trust in deployed systems. The field has converged on a near-consensus explanation: the language model component over-relies on learned co-occurrence statistics (language priors), producing objects that are likely in the described scene even when they are absent from the image. This paper challenges that consensus. Using cosine similarity analysis and Smooth Grad-CAM entropy, the authors show that 38–52% of object hallucinations across three MLLM architectures originate not in the language prior but in failures of the visual encoder itself.

AT A GLANCE

Problem: Object hallucination in MLLMs is blamed on language priors; visual encoding failures are ignored.

Why it matters: Misdiagnosing the root cause leads to interventions (decoding fixes) that do not address visual failures.

Method: Cosine similarity and Smooth Grad-CAM entropy to classify hallucinations as visual-origin vs. language-prior-origin.

Result: 38–52% of hallucinations are visual-origin; targeted visual fixes reduce POPE hallucination rate by 4.1 points and MSCOCO-Hallu rate by 37% relative.

Evidence: Three MLLM families; POPE, MME, MSCOCO-Hallu benchmarks; ablations of visual contrastive training and attention focusing.

The Language Prior Hypothesis and Its Limits

The dominant account of object hallucination posits that MLLMs inherit strong language priors from pre-training: the language model has seen “refrigerator” co-occurring

with “kitchen” thousands of times and will generate it even when the image shows an empty kitchen. Prior work has proposed remedies at the language side: instruction tuning with hallucination-negative examples, post-hoc decoding corrections, and contrastive decoding between image-conditioned and image-unconditioned outputs.

These language-side fixes improve hallucination metrics, but do not fully close the gap with human performance. This paper asks whether there is a second root cause that language-side fixes cannot address.

Diagnosing Visual-Origin Hallucination

The authors propose two diagnostic metrics that distinguish visual-origin from language-prior-origin hallucination.

Image-text cosine similarity measures how well the visual embedding aligns with the text of the generated output:

$$\text{sim}(v, t) = \frac{\mathbf{f}_v \cdot \mathbf{f}_t}{\|\mathbf{f}_v\| \|\mathbf{f}_t\|}$$

If the visual encoder correctly represents the image content, $\text{sim}(v, t)$ should be high for faithful outputs. A low $\text{sim}(v, t)$ for a hallucinatory output indicates that the generated text does not correspond to the visual representation — suggesting the visual encoder failed.

Smooth Grad-CAM entropy measures the diffuseness of image regions attended during generation:

$$H_{\text{Grad-CAM}} = - \sum_r p_r \log p_r$$

where p_r is the normalized Smooth Grad-CAM activation for image region r . High entropy means the model fails to focus on task-relevant objects. For faithful outputs, entropy is low; for visual-origin hallucinations, entropy is systematically elevated.

Using these two metrics as a joint classifier, the authors assign each hallucinated object instance to one of three categories:

- **Language-prior-origin:** high sim, low entropy, object absent from image.
- **Visual-origin:** low sim or high entropy, regardless of whether the object could be predicted from language priors.
- **Ambiguous:** both factors elevated.

Quantitative Decomposition

Across three architectures (LLaVA-1.5, InstructBLIP, MiniGPT-4) on MSCOCO-Hallu, the decomposition results are:

Model	Language	Visual	Ambig.
LLaVA-1.5	41%	44%	15%
InstructBLIP	38%	47%	15%
MiniGPT-4	45%	40%	15%
Average	41%	44%	15%

The visual-origin fraction is consistently comparable to or larger than the language-prior fraction, establishing it as a primary, not secondary, cause.

Targeted Remediation

The authors propose two complementary fixes targeting the visual encoder.

Visual contrastive training pushes image embeddings toward representations of faithful captions and away from unfaithful ones:

$$\mathcal{L}_{\text{contrast}} = -\log \frac{\exp(\text{sim}(v, t^+)/\tau)}{\exp(\text{sim}(v, t^+)/\tau) + \sum_j \exp(\text{sim}(v, t_j^-)/\tau)}$$

where t^+ is the ground-truth caption and t_j^- are negative captions (hallucination-containing descriptions generated by the model).

Attention focusing regularization reduces visual entropy during training by penalizing diffuse visual attention:

$$\mathcal{L}_{\text{focus}} = \lambda_2 \cdot H_{\text{Grad-CAM}}$$

The full training objective combines both:

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda_1 \mathcal{L}_{\text{contrast}} + \lambda_2 \mathcal{L}_{\text{focus}}$$

Experimental Results

Model	POPE Acc.	Hallu. Rate
LLaVA-1.5 (baseline)	85.3%	28.4%
+ Language-side fix (ICD)	87.1%	24.9%
+ Visual contrastive training	88.6%	20.1%
+ Full method	89.4%	17.8%

The full method reduces hallucination rate by 37.3% relative on MSCOCO-Hallu and improves POPE accuracy by 4.1 points, without degrading object recognition or VQA performance.

Ablations and Interpretation

Contrastive vs. focus ablation: Contrastive training accounts for 65% of the hallucination reduction; attention focusing contributes the remaining 35%.

Cross-architecture AUC: The diagnostic metrics predict visual-origin hallucinations with AUC 0.78–0.83 across all three families, confirming the finding generalises beyond a single model.

No regression on downstream tasks: VQAv2 and ScienceQA scores are unchanged, confirming the intervention is targeted and does not erode general multimodal capability.

Diagnostic threshold sensitivity: The 41–44% visual-origin estimate shifts by at most ± 6 points across threshold choices, showing the result is robust to the classification boundary.

Reference

Peiyang Xu, Xiaopei Zhu, Jun Zhu, and Xiaolin Hu. **Beyond Language Priors: Diagnosing and Fixing Visual-Origin Hallucinations in Multimodal LLM.** arXiv:2609.00231, August 2026. <https://arxiv.org/abs/2609.00231>